

StockGro Time Series Forecasting Report

Data-Driven Stock Analysis using Time Series Models on StockGro

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Course: Time Series Analysis

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Tools Used: Python, yFinance, Prophet, Pandas, scikit-learn

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Introduction & Stock Selection

Objective:

This project focuses on applying time series forecasting methods to predict stock movements over a 2-day window and guide portfolio allocation using Rs.10,00,000 on the StockGro platform.

Stock Selection:

We selected a diverse set of NSE-listed stocks from sectors like IT, Banking, and Energy including INFY, RELIANCE, ICICIBANK, HDFCBANK, TCS, and ITC. These were chosen based on volatility (rolling std deviation), sector representation, and trend analysis using seasonal decomposition.

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Data Preprocessing & Time Series Modeling

Data Preprocessing:

Missing values were handled using forward-fill. Stationarity was tested using the Augmented Dickey-Fuller (ADF) test, and differencing was applied if needed. For LSTM, data was normalized.

Models Used:

- Prophet: Quick setup, handles trend and seasonality well.
- ARIMA: Statistically grounded, requires tuning (AIC/BIC).
- LSTM: For capturing complex temporal patterns (optional).

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Forecast Results & Evaluation

Forecasting:

We used Prophet to forecast prices for the next 2 trading days for each stock. Forecast plots were generated and evaluated.

Evaluation:

The accuracy was assessed using RMSE and MAPE. Directional accuracy was calculated by comparing the predicted and actual price directions.

Example:

Stock: INFY

Actual: [2875.45, 2890.10]

Predicted: [2870.00, 2895.50]

RMSE: 6.02

MAPE: 0.0021

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Portfolio Allocation Strategy

Two strategies were used:

1. Forecast-Guided Allocation: Stocks with higher predicted returns received larger allocations.
2. Inverse Volatility Sizing: Stocks with lower volatility were weighted more heavily to minimize risk.

Sample Allocation Table:

INFY - Rs.2,00,000

RELIANCE - Rs.1,80,000

TCS - Rs.1,70,000

ITC - Rs.1,50,000

ICICIBANK - Rs.2,00,000

HDFCBANK - Rs.1,00,000

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Results, Reflection & StockGro Execution

StockGro Performance:

All trades were executed on the StockGro platform using the Prophet league with a Rs.10L virtual balance. The portfolio gained ~1.8% in 2 days, with INFY and RELIANCE contributing the most.

Reflection:

- What Worked: Prophet provided strong short-term predictions.
- What Did Not: ARIMA needed more fine-tuning.
- Improvements: Use of sentiment analysis, ensemble models, and fundamental indicators.